AI-Assisted PR Review Pipeline

A governed, multi-agent system that helps engineering teams review pull requests faster and more consistently

THE PROBLEM

Engineering teams spend significant time on manual pull request review — and that time is inconsistent. In a controlled review pass, human reviewers correctly identified real issues but were inconsistent about explaining how severe each issue was or what action to take, scoring only 1 out of 2 on that specific dimension across every case tested. Review capacity is also a bottleneck: at roughly 5 minutes and an estimated $6.25 in reviewer time per pull request, this adds up quickly at team scale.

THE SOLUTION

A pipeline of specialized AI agents — a planner, a reviewer, and a release-note writer — work together under strict, enforced permission boundaries. The reviewer agent reads each change, flags real issues with clear severity and a recommended action, and automatically escalates anything risky to a human instead of approving it. No agent can merge code, touch credentials, or act outside its assigned role — this is enforced in the underlying system, not just a written policy.

EARLY RESULTS

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known issues correctly caught in initial testing	issues caught in held-out test cases	false alarms raised on clean, issue-free changes	automated safety checks run before every change ships

Based on an initial evaluation using representative public code changes with known, intentionally introduced issues — not yet run on this team's live repository. See the full impact study for methodology.

WHY IT'S SAFE TO ADOPT

- Every AI agent only has access to the specific tools and data it needs for its job — nothing more.
- Any risky or ambiguous change is automatically routed to a human, never auto-approved.
- A permission-drift test was run deliberately: a real misconfiguration was introduced, and the system's automated checks caught it and blocked it before it could reach production, exactly as designed.
- No AI agent can merge code, access secrets, or bypass a human on high-risk changes.

WHAT WE'D NEED TO GO LIVE

This system is proven on representative test data. To pilot it on our own codebase, we'd need: read access to a real (or sanitized) repository, a small budget for AI API usage (early estimates suggest well under $10/month at typical review volume), and a 2-4 week pilot window with a human still reviewing every AI-flagged change before it merges.

Full technical documentation, architecture details, and evaluation methodology available on request.